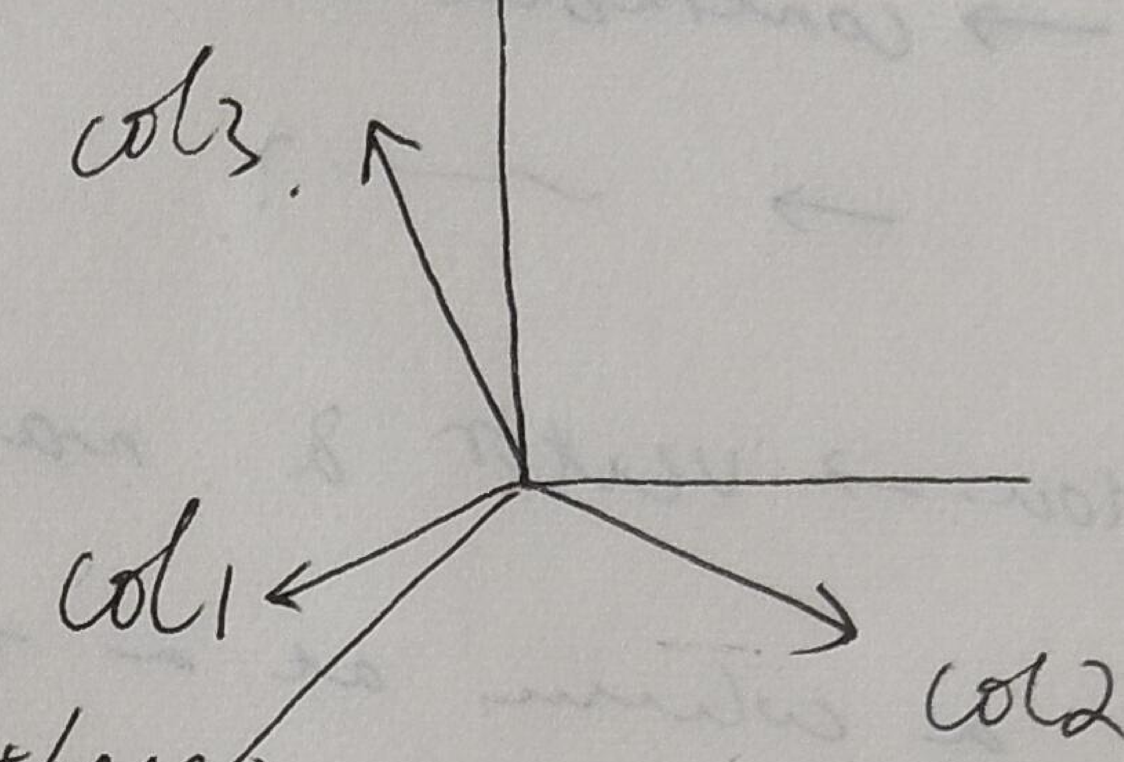
$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -3 & 4 \end{bmatrix} \quad b = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}$$

Row picture → plane (a point)
[solution]

⇓

quit.

$$x\begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + y\begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} + z\begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}$$



Solution: $x=0$. $y=0$. $z=1$.

(just <u>want</u> "z" (one of these)

Thus. (0,0,1). → otherwise, systematic way. Elimination)

$$x\begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + y\begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} + z\begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix}$$

Thought process: "took one of these and one of those".

(cook it). $x=1$ $y=1$. $z=0$. (notice that..)

Solve $Ax=b$ for every b? <u>OR</u>.

★Do the linear combs. of the The columns fill 3-D space?

A good matrix → A non-singular matrix
An invertible matrix.

3 columns all lie in the same plane

trouble: the combinations → the b is out of the plane
and UNREACHABLE.

↓

singular. / not invertible.

→ 9 dimensions?. Every b?. ! may very well fill out the whole 9-D. space

→ any / random. dimensions.

→. sometimes _ . all beautiful.

→. NOT independent.

→ contributes nothing new. ← but if.

→ ~ . ?

How. → vector & . matrix. multiplication.

!. a column at a time.

$A x = b.$

$$\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1\begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2\begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 12 \\ 7 \end{bmatrix}.$$

the Other way of thinking (also "dot. product").

2 5 1 2. $2\times1+5\times2$ is the 12.

1 3 1 2. 7.

☆ 1. by columns.

2. OR by rows.

// x → dot product.

☆ Thought: A x is a comb of columns of A.

—— MIT. 18.06. Lecture 1.

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